Software typical operation steps

一、Software installation：

1. Double click the tqECU Pro.exe file and install the computer software by step。
2. Double click the ftdi_ft232_drive.exe file and install the FT323 driver ,Enter key when you jump out of the prompt。

二、Hardware connection：

1. The communication line is connected to the ECU, and the ECU is connected to the motorcycle.
2. Turn on the main switch of the motorcycle.
3. Connect the communication line to the computer USB interface.
4. open ECU PRO computer software.
5. Click

switch,Display the ON and the green background represent communication connection normal.
6. If can't connect, please choose menu 【**Select Com**】 and change a communication port to try again.

三、Data processing：

1. Click

,read the data from the ECU。
2. Start the motorcycle test and adjust the fuel injection, ignition angle and other parameters according to the needs。
3. Click

,write data to ECU when the motorcycle is not started。
4. Start the motorcycle and test the modified data。
5. Click

or

to save the modified data 。
6. Click

reset to Factory Defaults if needed。

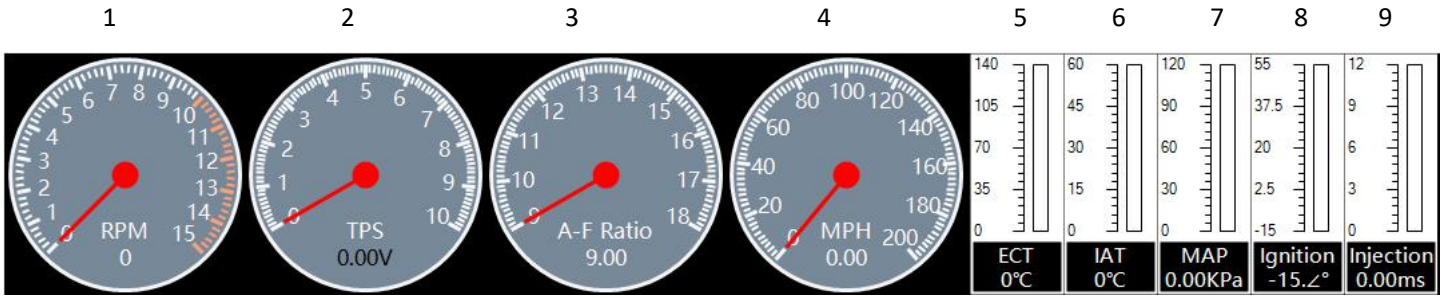
四、How to study TPS voltage：

1. Click【Study TPS】when hardware connection is normal。
2. The software jump out of the prompt box, TPS from the minimum to the maximum continuous operate three times.
3. After study TPS success, the prompt box will disappear itself, and the minimum and maximum voltage will be automatically saved in 【TPS Voltage(min.)】and【TPS Voltage(max.)】 form of options.

五、Shortcut key description：

F1：Selected table data +1
F2：Selected table data -1
F3：Selected table data +0.1
F4：Selected table data -0.1
I ：Selected table data automatic averaging

六、Display instructions：



1. RPM
2. TPS
3. Air-Fuel Ratio
4. Speed
5. Engine temperature
6. Intake temperature
7. Intake pressure
8. Ignition angle
9. injection time

六、How to adjust the idle RPM：

1. Adjusting idle RPM by fuel injection time. In general, only adjust the fuel injection rate of 1000 to 2500 of the IDLE table.According to the actual RPM adjustment.Too much or too little fuel injection will affect the idle RPM.For example, the normal idle RPM is 1500. If the RPM rises to more than 2000, the amount of fuel injection of 2000 to 2500 can be reduced.

Injection Oil	Injection Angle	Ignition Angle	Ignition Time				
TPS	500	1000	1500	2000	2500	3000	3500
100%	0	0	0	0	0	0	0
90%	0	0	0	0	0	0	0
80%	0	0	0	0	0	0	0
70%	0	0	0	0	0	0	0
60%	0	0	0	0	0	0	0
50%	0	0	0	0	0	0	0
40%	0	0	0	0	0	0	0
30%	0	0	0	0	0	0	0
20%	0	0	0	0	0	0	0
10%	0	0	0	0	0	0	0
0%	0	0	0	0	0	0	0
IDLE	0	0	0	0	0	0	0

2. Adjusting idle RPM by ignition angle. In general, only adjust the ignition angle rate of 1000 to 2500 of the IDLE table.According to the actual RPM adjustment.In normal conditions, the higher the angle, the higher the RPM.For example, the normal idle RPM is 1500, if the RPM rises to more than 2000, the angle of 2000 to 2500 can be reduced.

Injection Oil | Injection Angle | **Ignition Angle** | Ignition Time

TPS	500	1000	1500	2000	2500	3000	3500
100%	0	0	0	0	0	0	0
90%	0	0	0	0	0	0	0
80%	0	0	0	0	0	0	0
70%	0	0	0	0	0	0	0
60%	0	0	0	0	0	0	0
50%	0	0	0	0	0	0	0
40%	0	0	0	0	0	0	0
30%	0	0	0	0	0	0	0
20%	0	0	0	0	0	0	0
10%	0	0	0	0	0	0	0
0%	0	0	0	0	0	0	0
IDLE	0	0	0	0	0	0	0

3. Adjust idle RPM by Idle Motor(Note: it is only apply with Idle Motor)， The Idle Motor data can be adjusted according to the engine temperature.The Idle Motor data too large or too small will affect the RPM stability.For example, , if the idle RPM is too low ,you can modify the data corresponding to the engine temperature 98℃when the engine temperature is 98℃.

Ignition Angle | Ignition Time | Air-Fuel Ratio | Comp Temp | Comp Map | **Idle Motor**

	126℃	112℃	98℃	84℃	70℃
	73	73	75	75	77

4. Modifying the special parameters to adjust the stability of idle RPM。

①. Modifying the Idle Speed(Cold) and the Idle Speed(Hot)

Comp Map | Idle Motor | Setup Air-Fuel | **Options** | Setup

Idle Speed (Cold)	1600	RPM
Idle Speed (Hot)	1400	RPM

②. Modifying Minimum Injection(Cold) and Modifying Minimum Injection(Hot)（Usually not required to modify）

Comp Map | Idle Motor | Setup Air-Fuel | **Options** | Setup

Minimum Injection (Cold)	3.40	ms
Minimum Injection (Hot)	2.50	ms

③. Modifying Maximum Idle Angle Reduction(Cold and Hot) and Idle angle increased(Cold and Hot)， （Usually not required to modify）

Comp Map | Idle Motor | Setup Air-Fuel | **Options** | Setup

Maximum Idle Angle Reduction(Cold)	-10	∠°
Maximum Idle Angle Reduction (Hot)	-10	∠°
Idle angle increased (Cold)	10.0	∠°
Idle angle increased (Hot)	10.0	∠°

七、How to modify the speed up no good

1. Testing under injection oil table， In the coordinate system composed of TPS and RPM, the fuel injection data is modified according to the path of the color box. The data can be modified according to the test result of the air fuel ratio tester.
2. Testing under ignition angle table， In the coordinate system composed of TPS and RPM, the ignition angle data is modified according to the path of the color box.The ignition angle can be judged by the sound of the engine.High engine sound, knock, knocking and so on are too high angle.The engine sound is dull, the gun, the exhaust smoke, the exhaust pipe heat and so on are too low angle.
3. The temperature compensation of fuel injection can be modified under 【Comp Temp】table.

八、How to adjust the fuel injection time.

1. Connect the ECU according to the document （二、**Hardware connection**）.
2. Click Reader read data from ECU.
3. Click Injection Oil.
4. Select the form to modify the data, can directly enter the data or through the shortcut key
5. Click Writer write data to ECU.
6. Start the engine test,
7. Click Save or Save AS to save the modified data.

Note： the same of the steps of adjusting the ignition angle

九、How to modify the air fuel ratio

1. Connect the ECU according to the document （二、**Hardware connection**）.
2. Click Reader read data from ECU.
3. Click Setup Air-Fuel， setting the corresponding value of the air fuel ratio and the voltage of the oxygen sensor。
4. Click Air-Fuel Ratio ,the air fuel ratio control value is added, and the table of value 0 means that does not control the air fuel ratio.(Note: the ECU will not handle the air fuel ratio control when the "Don't Air-fuel Ratio" is checked under the "setup" interface.)
5. Click Writer write data to ECU.
6. Click Save or Save AS to save the modified data.

十、 How to modify the fuel injection compensation corresponding to the engine temperature

1. Connect the ECU according to the document （二、**Hardware connection**）.

2. Click

read data from ECU.

3. Click

.

4. The corresponding percentage compensation is adjusted according to the temperature of the engine.For example, if the fuel injection time (TPS:10%, RPM:4000) is 10ms,and (28 ℃, 4000) the compensation amount is 50%, so the fuel injection time is 10+10*50%=15 at 28 ℃(TPS:10%, RPM:4000).

5. Click

write data to ECU.

6. Click

or

to save the modified data.

十一、How to modify the injection compensation corresponding to the intake pressure

1. Connect the ECU according to the document （二、**Hardware connection**）.

2. Click

read data from ECU.

3. Click

.

4. Adjust the injection increment according to the intake pressure.For example, if the fuel injection time (TPS:10%, RPM:4000) is 10 ms ,and (72Kpa, 4000) the compensation amount is 1ms, so the fuel injection time at 72Kpa (TPS:10%, RPM:4000) is 10+1=11ms.

5. Click

write data to ECU.

6. Click

or

to save the modified data.